Software Engineering: Algorithms and Data Structures
Example 1
MARK: 92%

```
#include<iostream>
#include<string>

using namespace std;

struct Node{
        string data;
        Node* next;
};

typedef Node* NodePtr;

//Functions

void insmlistend(string data, NodePtr &hdlist){
        NodePtr searchPtr, lastPtr, newPtr;
        newPtr = new Node;
        newPtr -> data = data;
        newPtr -> next = NULL;
        if(hdlist==NULL){ // insertion into empty list
                hdlist = newPtr;
                return;
        }
        else{
                searchPtr = hdlist;
                lastPtr = hdlist;
                while(searchPtr!=NULL){
                        lastPtr = searchPtr;
                        searchPtr = searchPtr->next;
                }
                newPtr->next=searchPtr;
                lastPtr->next=newPtr;
        }


};

bool ismlistsorted(NodePtr hdlist){
        bool sorted;
        NodePtr nextPtr;
        /// there is no protection against empty list
        nextPtr = hdlist->next;
```

```
        if(hdlist->data < nextPtr->data){
                sorted = true;
        }
        else {
                return false;
        }
        if(nextPtr->next!=NULL){
                sorted = ismlistsorted(nextPtr);
        }
        return sorted;
};


void printmlist(NodePtr hdlist){
        while(hdlist!=NULL){
                cout<<hdlist->data<<endl;
                hdlist=hdlist->next;
        }
};

void swapfirstlastmlist(NodePtr &hdlist){
        NodePtr searchPtr, lastPtr, tempPtr;
        tempPtr =hdlist;
        searchPtr = hdlist;
        lastPtr = hdlist;
        while(searchPtr->next!=NULL){
                lastPtr = searchPtr;
                searchPtr = searchPtr->next;
        }
        //At this point searchPtr points to the last item, lastPtr the penultimate, hdlist the
first, and tempPtr the second
        searchPtr->next =tempPtr->next;
        hdlist=searchPtr;
        lastPtr->next = tempPtr;
        tempPtr->next = NULL;

};


int main(){
        NodePtr hdlist = NULL;
        bool get_elements = true;
        while(get_elements){
                cout<<"Enter element"<<endl;
                string input;
                cin>>input;
                if(input=="STOP"){
```

```
                get_elements = false;
            }
            else{
                insmlistend(input, hdlist);
            }
        }
        cout<<endl;
        cout<<"Printing the list"<<endl;
        printmlist(hdlist);
        cout<<endl;
        cout<<"Printing whether the list provided in input is sorted or not:"<<endl;
        if(ismlistsorted(hdlist)){
            cout<<"The list is sorted"<<endl<<endl;
        }
        else{
            cout<<"The list is not sorted"<<endl<<endl;
        }
        cout<<"Printing the list after swapping the first and last node"<<endl;
        swapfirstlastmlist(hdlist);
        printmlist(hdlist);
        return 0;
}
```